

# SALES ANALYSIS PROJECT

DONE IN WEEK 4 OF MY PYTHON JOURNEY

## Abstract

All thanks to God Almighty for the strength and grace for this project; special thanks to my parents for providing the resources to register for the 2026 Matrix Cohort. Also, I extend gratitude to my tutor, the person of Sir Ginikachukwu Edward.

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# INTRODUCTION AND DATA PREPARATION

## Project Objective:

This report aims to analyse historical sales data to evaluate overall business performance, identify key revenue drivers, and provide actionable, evidence-based recommendations to management.

**Dataset Overview:** The raw dataset provided for this analysis contained 5,200 records detailing product sales, transaction quantities, pricing, regional distribution, customer order statuses, and payment methods.

**Data Cleaning and Preparation:** To ensure the integrity and accuracy of the analysis, the dataset underwent a rigorous cleaning process using Python's Pandas library:

- **Duplicate Removal:** 21 completely identical records were removed to prevent artificial revenue inflation.
- **Missing Value Handling:** Records missing critical mathematical data (price, quantity, or order date) were dropped. Conversely, missing categorical values (such as region or payment method) were filled with 'Unknown' to preserve the maximum amount of usable data.
- **Data Standardisation:** Text formats across payment methods and order statuses were stripped of whitespace and properly capitalised. Quantity column strings (e.g., 'pcs') were removed to allow for numerical conversion.
- **Derived Metrics:** A total\_sales column was calculated by multiplying the cleaned quantity and price columns to enable direct revenue analysis.

## Code Snippet:

Python

```
categorical_cols = ['product', 'region', 'customer_id', 'payment_method', 'status']
```

```
df[categorical_cols] = df[categorical_cols].fillna('Unknown')
```

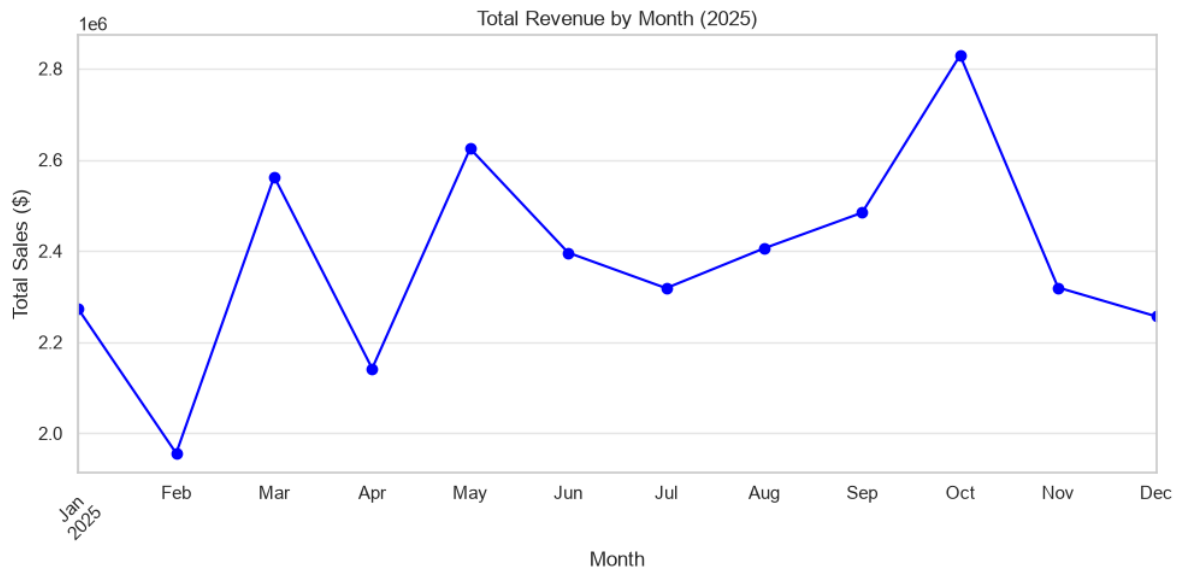
```
df.dropna(subset=['order_date', 'qty', 'price'], inplace=True)
```

After processing, the final cleaned dataset resulted in 4,480 highly reliable records ready for exploratory data analysis.

## REVENUE TRENDS & PAYMENT PREFERENCES

### Monthly Sales Performance

Tracking revenue over time is critical for understanding business momentum and seasonality.



*Fig1.0 Total Revenue by Month chart.*

**Interpretation:** The Monthly Sales Trend chart reveals the revenue progression throughout 2025, highlighting a massive peak sales period in October. The business generated \$2,828,990 in this single month, vastly outperforming the rest of the year. Management should investigate the specific drivers of this Q4 spike to replicate it in future campaigns.

### Payment Method Breakdown

Understanding how customers prefer to pay dictates where the business should invest its financial infrastructure.

[INSERT CHART: charts/revenue\_by\_payment.png HERE]

**Interpretation:** Direct bank transfers are the undisputed dominant payment method, generating significantly more gross revenue than cash or POS systems. The customer base heavily prefers digital, cashless transactions.

*Code Snippet:*

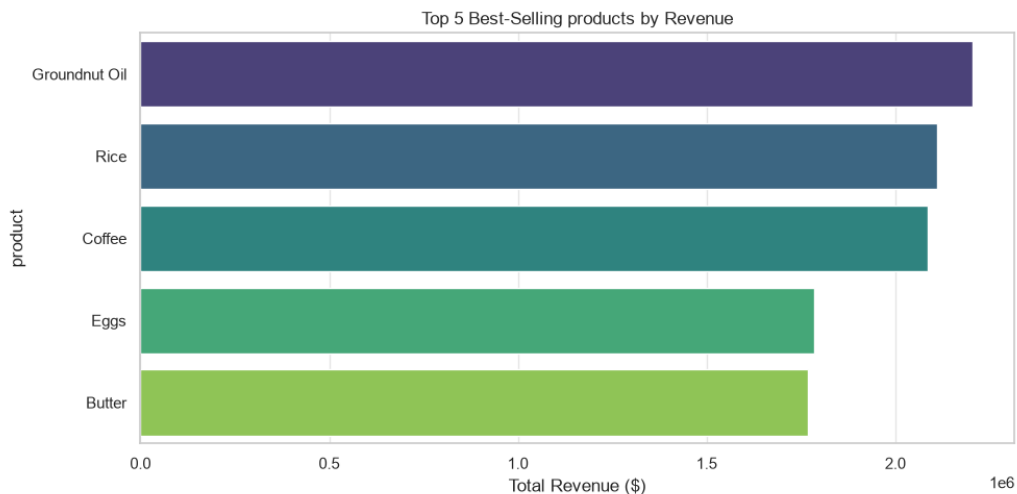
### Python

```
payment_revenue =  
df.groupby('payment_method')['total_sales'].sum().sort_values(ascending=False)
```

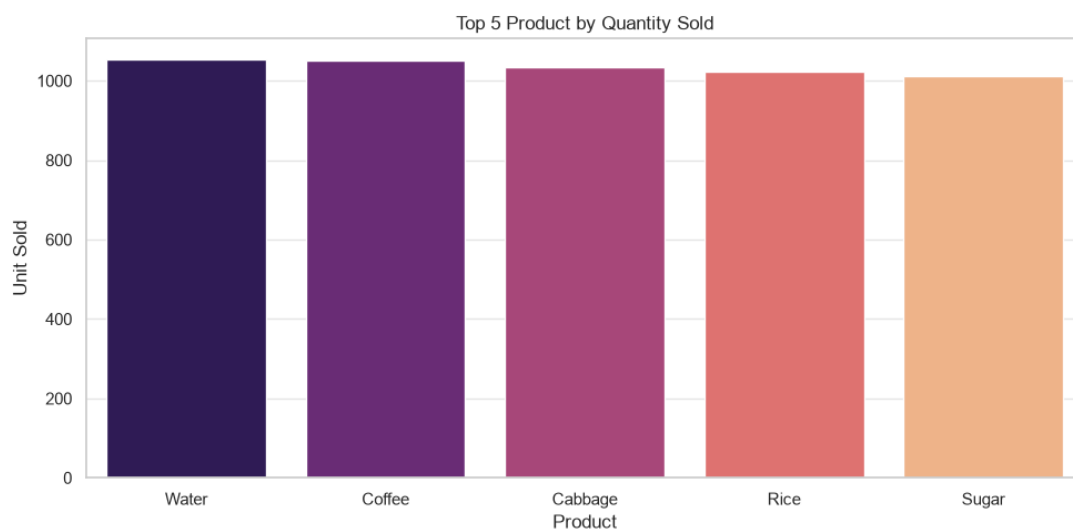
## PRODUCT PERFORMANCE

### Revenue vs. Volume Disconnect

A core finding of this analysis is the disconnect between the products that drive gross revenue and the products that drive transaction volume.



*Fig2.0 Top product revenue table.*



*Fig 2.1 Top product by Quantity sold.*

### Interpretation:

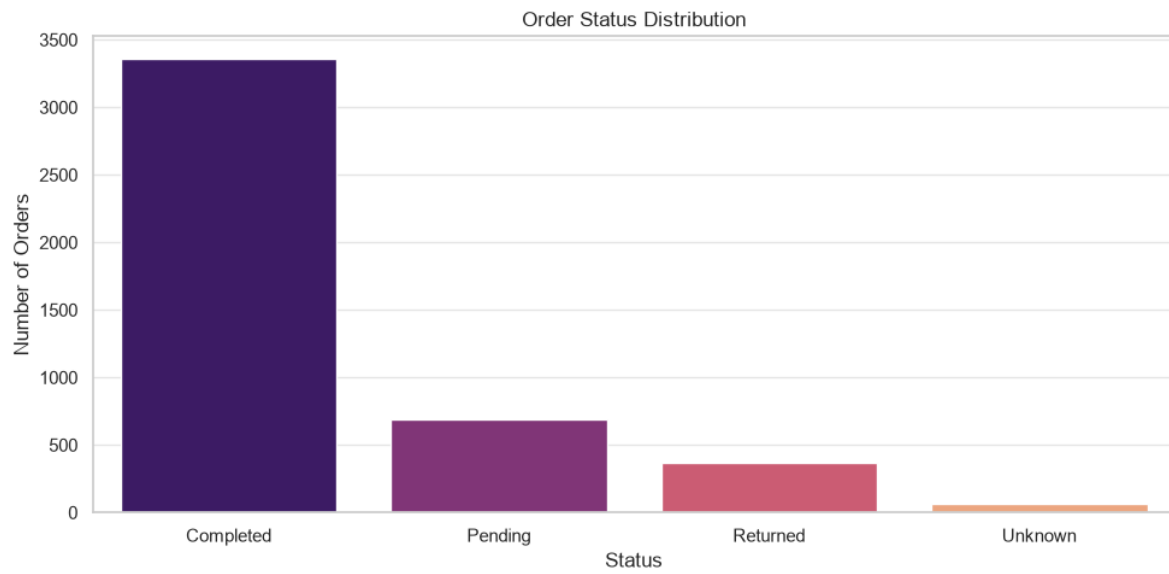
Groundnut Oil and Rice drive the most revenue, making them the most valuable items for the bottom line. However, everyday low-cost essentials like Water and Coffee drive the highest sheer volume of units sold.

**This presents a strategic opportunity:** high-volume items like Water bring customers through the door, while Groundnut Oil generates the actual profit. Marketing teams should consider bundling these high-volume items with high-margin items to maximise the cart size of every customer visit.

## OPERATIONAL HEALTH & FULFILLMENT

### Order Status Distribution

Protecting revenue requires ensuring that completed sales are successfully fulfilled without friction.



*Fig. 3.0 Order Status Distribution.*

### Interpretation:

The vast majority of orders were completed, indicating a generally healthy fulfilment process. However, 340 orders were classified as 'Returned'. While the completion rate is high, those returns represent lost revenue and wasted logistics costs. Operations should investigate the root cause—such as shipping damage, defective products, or incorrect fulfilment—to plug this revenue leak.

*Code Snippet: Analysing Order Status Frequencies*

### Python

```
status_counts = df['status'].value_counts()
print(status_counts)
```

## KEY INSIGHTS, RECOMMENDATIONS, AND CONCLUSION

### Key Business Insights

1. **Q4 Seasonality Spike:** The dataset reveals a massive revenue peak in October (\$2,828,990), indicating heavy seasonal demand that drastically outweighs the rest of the year.
2. **Volume vs. Revenue Disconnect:** Low-cost items like Water and Coffee drive the highest foot traffic and transaction volume, while Groundnut Oil and Rice are the primary drivers of gross revenue.
3. **Digital Payment Dominance:** Direct bank transfers accounted for over \$12 million in revenue, proving that the customer base heavily favors cashless, digital transactions over physical cash or POS.
4. **Return Rate Impact:** While the 3,327 completed orders show a healthy operational baseline, the 340 returned orders represent a measurable leak in sustained revenue and logistics costs.
5. **Customer Acquisition Baseline:** The established Average Order Value (AOV) of approximately \$6,376 provides a concrete metric to help set strict limits on future marketing and customer acquisition spend.

### Strategic Recommendations

- **Optimise Q4 Readiness:** Supply chain and human resources must prepare for the October demand spike well in advance. Securing additional inventory for high-revenue items (Groundnut Oil, Rice) and ensuring adequate fulfilment staffing before Q4 will prevent stockouts.
- **Implement Strategic Bundling:** Marketing should leverage the high transaction volume of everyday essentials (Water, Coffee) by bundling them with high-margin items. This will convert routine, low-value visits into higher-yield transactions.
- **Fortify Digital Infrastructure:** Given the overwhelming reliance on bank transfers, IT and operations must ensure the digital payment gateway has maximum uptime. Any technical failure in confirming transfers will immediately halt the company's primary revenue stream.
- **Audit the Fulfilment Pipeline:** Operations should conduct a targeted audit of the 340 returned orders to identify common denominators (e.g., specific carrier damage, defective batches, or delayed shipping times) and correct the root cause.

**Conclusion:** The historical sales data indicates a fundamentally healthy business with strong digital engagement and high-performing core products. By addressing the operational leak of returned orders and actively leveraging the distinct roles of high-volume versus high-revenue products, management is well-positioned to maximise profitability. Sustained focus on preparing for the October seasonal peak and maintaining robust digital payment infrastructure will be critical for scaling future revenue.